



Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL



Nigel Li^{1,2}

Yutong Han²

¹University of New South Wales, Sydney

²Columbia University, New York City

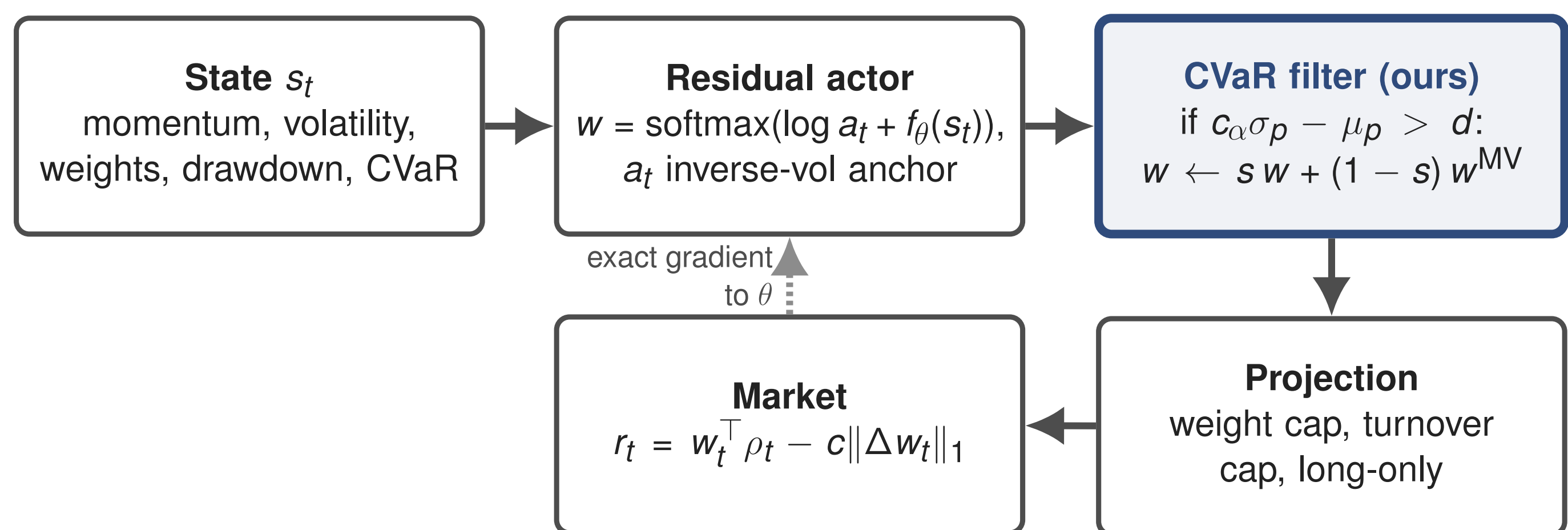
Abstract

A portfolio allocator trained by reinforcement learning (RL) to maximise return has no reason to avoid large losses. The usual safeguard caps Conditional Value-at-Risk (CVaR), the average loss in the worst weeks, through a Lagrangian penalty. We show that this safeguard can fail silently in two ways. When the penalty and the reward sit on different scales, the penalty weight stays at zero and the budget is never enforced, even as tail metrics improve. And the evaluation protocol can decide the winner, since a single stress window and a rolling walk-forward rank the same methods in opposite orders. We add a decision-time filter that moves each portfolio toward minimum variance when its forecast CVaR exceeds the budget. It cuts the out-of-sample breach rate from **75.1% to 7.4%**, and across a 40-fold, two-universe walk-forward the tail-risk reduction is significant (**26 of 27** untied folds, $p = 8.8 \times 10^{-6}$).

Problem Setup

Each week the agent splits capital across seven macro ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL), long-only with max weight 0.40, turnover cap 0.50 and a 5bps proportional cost. The data are 887 weekly returns from 2008 to 2024. $\text{CVaR}_{0.95}$ is the mean loss in the worst 5% of weeks, and the budget is $d = 1.2\%$. The policy is trained by back-propagating through the portfolio return itself,

$$\mathcal{L} = -\text{Sharpe} + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad \lambda \leftarrow \max\{0, \lambda + \eta_\lambda(\widehat{\text{CVaR}} - d)\}.$$



The filter forecasts CVaR from the trailing 104 weeks ($c_{0.95} = 2.06$) and keeps the largest s that clears d . It uses only data up to $t - 1$, and a shifted-returns test confirms no look-ahead.

Why a Tail Limit

In the week of 16 March 2020 the S&P 500 fell about 15% and the VIX closed at 82.69. An allocator trained only for return kept holding equities. An identical allocator with a CVaR limit had already rotated into Treasuries and cash, and lost about half as much (**7.7% vs 16.3%** drawdown).

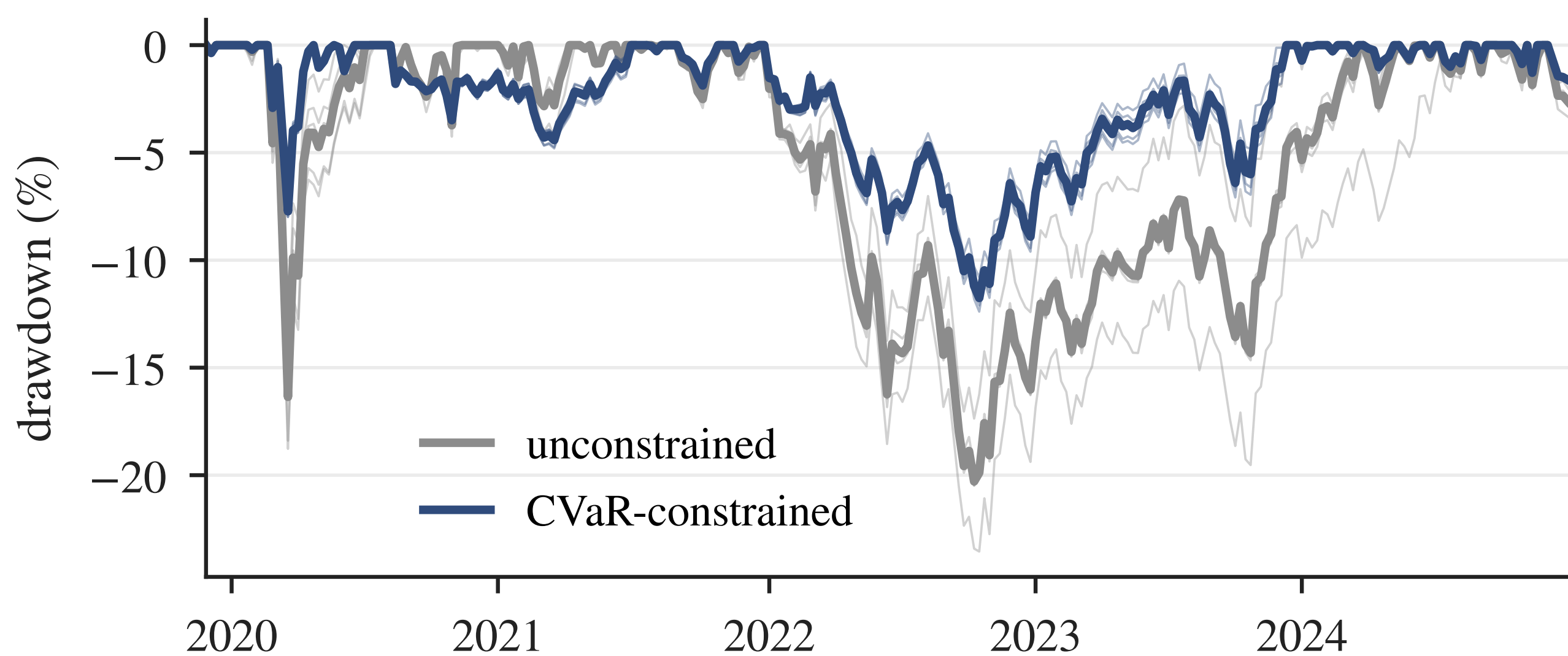


Figure 1. Drawdown on the 2020–2024 stress window (five-seed mean in bold).

Failure 1: The Penalty Never Switches On

The CVaR excess is about 10^{-2} . With a dual step $\eta_\lambda = 10^{-3}$, each update moves λ by about 10^{-5} , and 1500 updates are not enough. A model-free actor–critic fails the same way when its cost advantage is left at 10^{-3} against a standardised reward advantage.

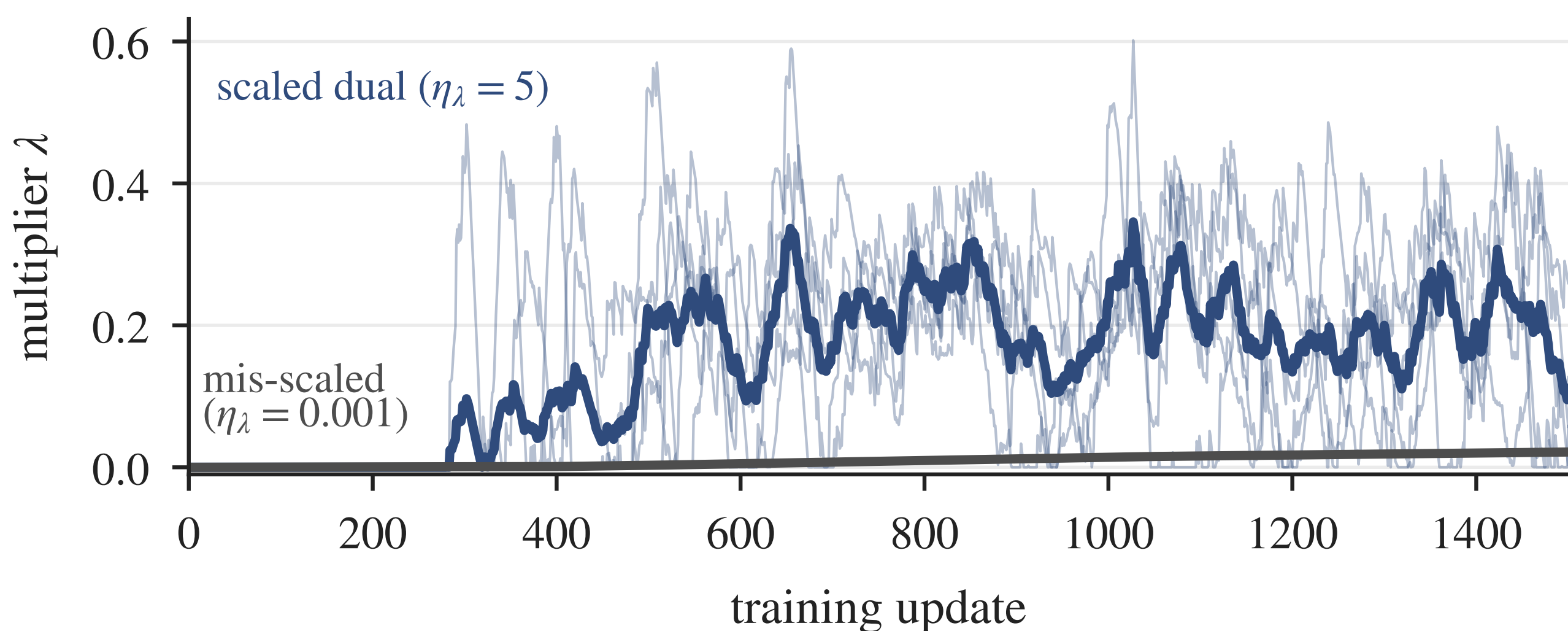


Figure 2. The Lagrange multiplier over training (five seeds, mean in bold).

The rest of the pipeline already cuts CVaR_{99} by **61%** with $\lambda \approx 0$ ($p = 0.0044$), so tail metrics improve whether or not the dual works.

☒ **Diagnostic.** Plot λ over training. If it stays at zero, the constraint is off.

Failure 2: The Test Protocol Picks the Winner

On a 31-asset universe, one stress split ranks the learner above every optimiser. A 20-fold rolling walk-forward reverses the ranking. Ledoit–Wolf shrinkage does not explain it. The tail reduction against unconstrained RL holds under both protocols ($p = 6 \times 10^{-4}$), while the edge over the optimiser does not.

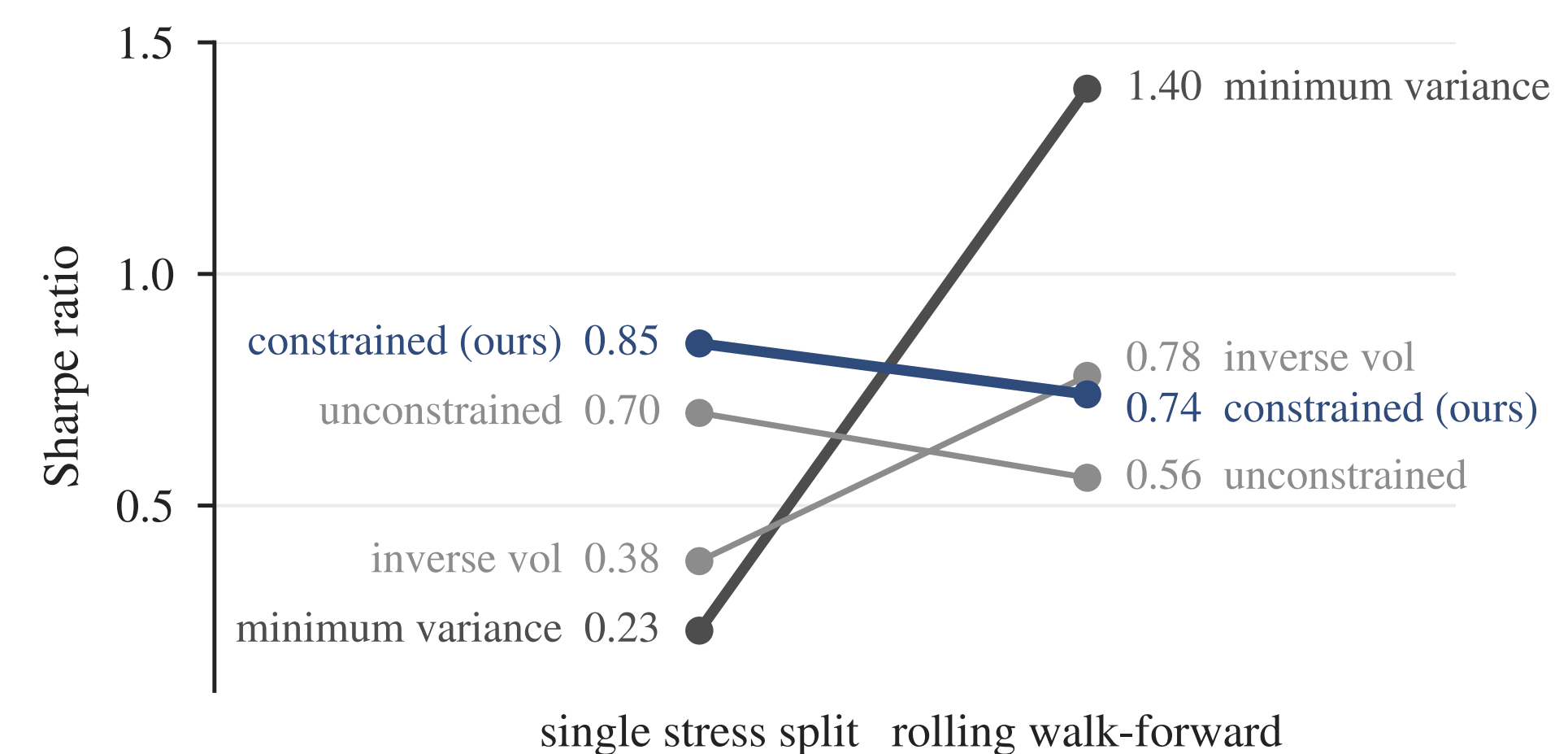


Figure 3. Sharpe ratio of the same methods under two protocols.

☒ **Diagnostic.** Rank learners against optimisers on rolling walk-forward folds, with Deflated Sharpe.

The Fix: A Decision-Time CVaR Filter

Scaling the dual fixes training but not the test window. The scaled dual binds in training and still breaches in 75% of test weeks, and four rules for setting d all give 71–78%. The filter acts on the portfolio itself, before it trades.

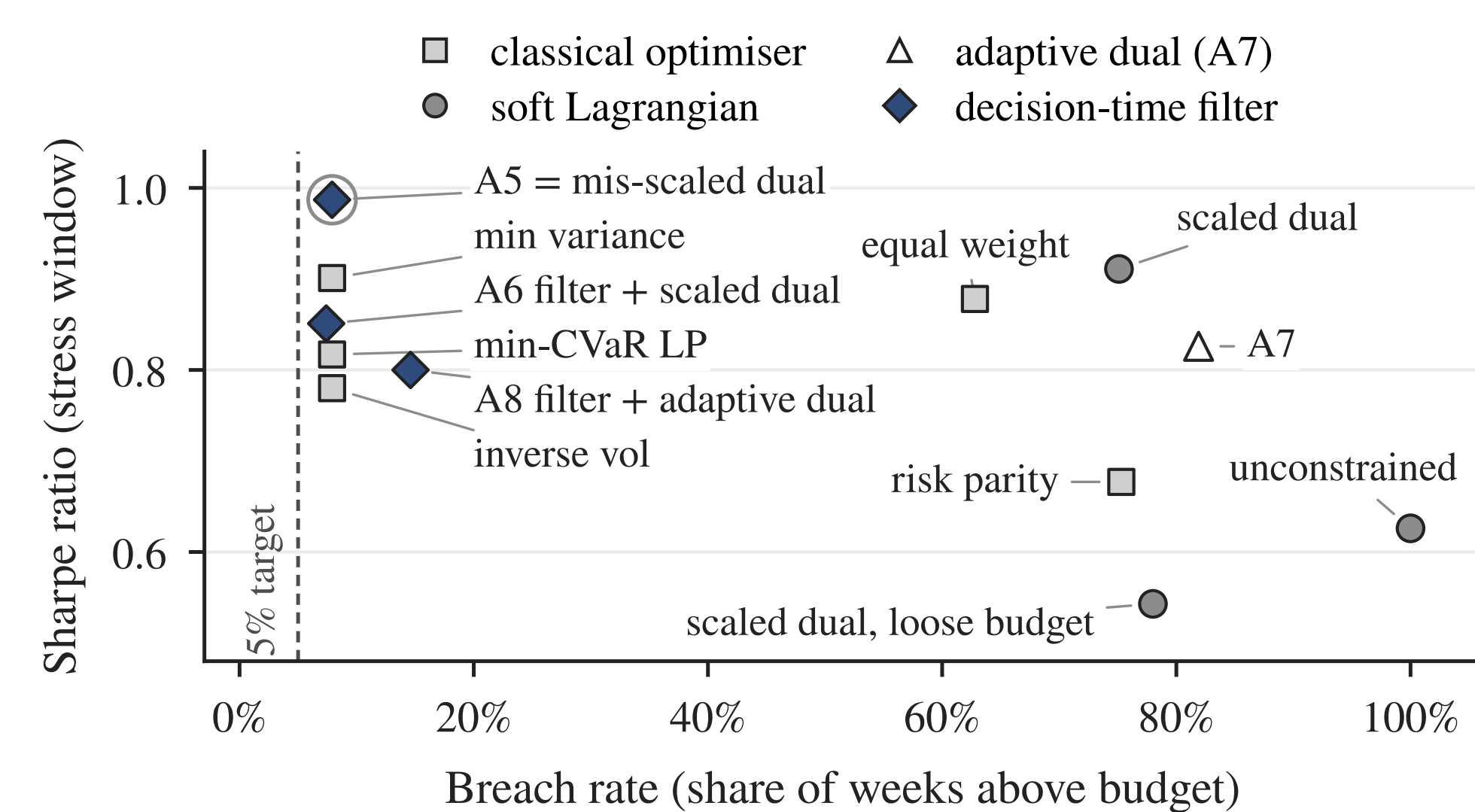


Figure 4. Breach rate against Sharpe on the stress window. Dashed line, 5% tolerance.

arm (5-seed mean)	Sharpe	CVaR ₉₉	breach
A1 unconstrained	0.626	0.065	100.0%
A2 mis-scaled dual	0.987	0.025	7.9%
A3 scaled dual	0.911	0.028	75.1%
A7 adaptive dual, no filter	0.826	0.031	81.9%
A6 filter + scaled dual	0.851	0.016	7.4%
minimum variance	0.901		7.9%

A7 raises λ online and still breaches, so the gain in A6 comes from the filter.

Statistical Significance

Rolling walk-forward with 26-week test folds on two universes, CVaR_{99} of the constrained vs the unconstrained learner.

universe	folds	improved	Wilcoxon p
macro ETF (7)	20	9 (11 tied)	0.0077
sector ETF (10)	20	17	0.0003
pooled	40	26 of 27 untied	8.8×10^{-6}
clustered by calendar	20		3.3×10^{-4}

Benjamini–Hochberg rejects the null on both universes. A tie is a fold where the budget never binds. On sector ETFs the stress result replicates, with CVaR_{99} down 30% at no Sharpe cost.

Limitations and Takeaways

The learner does not beat rolling minimum variance on Sharpe out of sample, and Deflated Sharpe removes its Sharpe edge. What survives correction is lower tail risk and budget compliance. The study uses weekly data and proportional costs, and the Gaussian forecast can understate fat tails.

☒ **Plot λ .** A flat trace at zero means the constraint is off.

☒ **Match the scales.** Standardise the cost advantage and size η_λ to the CVaR excess.

☒ **Test on rolling folds,** with Deflated Sharpe, before claiming an edge.

☒ **Enforce the budget at trade time.** A soft penalty only discourages breaches on average.

